

growtoo rebuild packet

Tech state · languages · data room · 3 September 2026 · internal, not a marketing page

This is the current production system at **https://growto.live**. The live app is a vanilla JS PWA-shaped site (no native store build, no service worker / web manifest in repo). Root README.md is stale — it still says “localStorage only, no backend.” Ignore it. Do not treat pitch-deck numbers, mint counts, or fundraising asks as product facts.

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1. Product snapshot

growtoo is a free cannabis / CBD *grow journal* with reminders, a 3-day weather widget, in-app camera, an AI coach (Gemini), photos, growth-stage tracking, public Stories, and an optional Solana Devnet proof layer (Tokenise + Market). Two desks after signup:

Grower

Journal is the product. Wallet is optional.
Tokenise (seal stages / seed NFT) and listing sit behind chainOptIn.

Adopter

Market / garden desk. Faucet and Invest are adopter-only. Journal plants are empty by default.

profileType is **immutable** after create (grower | adopter). Clients cannot flip it; Admin SDK can. Age gate is 18+. Product is a documentation tool, not legal advice. Follow the law where you grow.

Live site	https://growto.live
App	https://growto.live/app/
Sign-in	https://growto.live/dnevnik/

Public journal	https://journal.growto.live (also /journal/ on apex)
GitHub	mgebela/balkan-pharm-main
Firebase	balpha-9dab9 · Functions region europe-west1
Chain	Solana Devnet only — test \$GROWTOO has no monetary value

2. Live vs mocked vs planned

Live (no wallet)	Plants, journal, calendar, Log FAB, grow log, camera, illness-check helper, AI Coach (verified email for Gemini; local helper otherwise), reminders, 3-day weather, toolbox/charts, Stories + public journal, inbox, EN/HR/DE, Firebase auth (email + Google), theme (light default).
Live (Devnet opt-in)	Wallet link (Phantom / Solflare, ed25519 via linkwallet), stage seal / seed NFT queue, growth mints, quests gate, test \$GROWTOO rewards, activity bonus, rank/XP, market (14-day care trail + ≥50% care days, instant sale vs adopt-stake 50/50, on-chain escrow program), adopter garden, daily faucet.
Not live / mocked	Physical harvest redemption (specified, not a fulfillment pipeline). \$GROWTOO monetary value. Mainnet. Trustless attestation that a physical plant exists (on-chain = process evidence from the journal trail).
Locked to two emails	Admin panel and soil-moisture tool: supadmin@dnenvnik.live, admin@dnenvnik.live only. Firestore role is not enough.

3. Stack & hosting

Frontend	Static HTML + CSS design tokens + vanilla JS. No React/Vue/bundler for the app. Firebase JS SDK compat 12.16.0 from gstatic CDN. Sign-in page uses modular SDK.
CSS	styles/tokens.css (229 tokens, light via :root[data-theme=light]). App: app/styles/app.css.
Hosting	Netlify from main (publish ., no build). GitHub Pages is a secondary mirror only.
Backend	Firebase Auth, Firestore, Storage. Cloud Functions v2 (Node 22). GitHub Actions workers for Solana queues.
AI	Google Gemini gemini-2.0-flash via @google/genai. Secret: GEMINI_API_KEY.
Weather	WeatherAPI.com, 3-day forecast, city string (not GPS). API key is currently in client JS — move server-side on rewrite.
Wallets	Phantom / Solflare. Browser RPC is proxied: solanaRpc CF (QuickNode/Helius secret server-side).

On-chain program	Anchor 0.30.1, programs/growtoo-escrow, program id GspPo6doBKOYmD6aCFHgo2q3CEXmWEoZXPpXAJnkjdyb.
App Check	reCAPTCHA Enterprise site key live, monitor mode (APP_CHECK_ENFORCE off). Do not flip enforcement mid-session.

Monolith sizes (approx.): app/js/app.js 8.9k lines · plant-token.js 4.9k · market.js 3.1k · ai-coach.js 3.0k · app/index.html 1.4k.

4. Repo map

app/	signed-in product (index.html + js/ + styles/)
dnevnik/	sign-in / sign-up
journal/	public grower journal mini-site
index.html	marketing landing
locales/	EN / HR / DE dictionaries
js/	site chrome (i18n, appcheck, landing-market, trust)
styles/	tokens + landing CSS
functions/	Cloud Functions (Node 22)
chain/	Devnet workers, deploy scripts, traffic seed (gitignored keys/)
programs/	Anchor escrow program (Rust)
idl/	program IDL
firestore.rules	production rules
storage.rules	users/{uid}/journal/**
token-metadata/	\$GROWT00 + seed collection JSON/images
pitch/ risks/ rwa-docs/ terms/ privacy/ public papers	
community/ + community-agent/	social drafts (not product)
emails/	Firebase Auth email templates
docs/	this packet + test readiness (excluded from GitHub Pages, not from Netlify)

Local chain keys live in chain/keys/ (gitignored). Never commit service accounts or Solana keypairs.

5. Languages

Three live locales in locales/locales.json:

Code	UI name	Intl	URL
en	English (source of truth)	en-GB	/ (default)
hr	Hrvatski	hr-HR	/hr/* rewrite → same HTML
de	Deutsch	de-DE	/de/* rewrite → same HTML

- One HTML per page. English stays in markup / T('key', 'English') fallbacks. Missing translation shows English, never a key name.

- Marketing URLs are indexable per language. The app switches in place via `?lang= + localStorage['growtoo:lang']` and reloads (`data-i18n-reload`).
- Namespaces: `common`, `landing`, `app`, `signin`, `journal`, `terms`, `privacy`, `risks`, `rwadocs`, `pitch*`. Runtime: `js/i18n.js`.
- Croatian plurals use `_one` / `_few` / `_other`. Check: `npm run i18n:check`.
- Adding a language: `i18n-add-locale.mjs` + a Netlify rewrite pair. See `locales/README.md`.
- **Coach Gemini locale is only en or hr** in `coachChat`. German UI falls back to English for live coach copy. Illness-frame analysis uses English or Croatian the same way.
- Wallet-link challenge message is **not** translated (signature is byte-for-byte).
- Journal **data keys** stay Croatian internally: `stages` `klijanje` → `sadnica` → `vegetativna` → `cvjetanje` → `susenje`; care types `zalijevanje`, `gnojidba`. Labels are translated. Do not branch on displayed text.

6. Auth & identity

- Firebase Auth: email/password + Google. Authorized domains include `growto.live`.
- Signup writes `users/{uid}` with `role: 'user'` and `profileType`. Grower extra: `growSetup` (`indoor` / `outdoor` / `mixed`), `homeCity`. Adopter extra: `adopterIntent`, `acceptedDevnet`.
- Live Gemini requires `email_verified` on the ID token. Unverified accounts get a local helper. After admin-verifying an email, the user must sign out/in.
- `solanaPubkey` / `walletVerified` are written only by CF `linkWallet` (Admin SDK) after `ed25519` verify. Clients may unlink (delete those fields).
- `chainOptIn` is a user-profile flag for `Tokenise/Market`.
- Admin UI is email-allowlisted, not role-only.

7. Data room

Primary user journal is **one Firestore document**: `users/{uid}/app/state`. Local-first: same arrays also live in `localStorage` (`dnevnik-live-plants`, `-entries`, `-toolbox`). Debounced sync ~500ms. Hard ceiling: Firestore 1 MiB per document. Photos used to be inline base64 and broke backup; they now go to Storage.

7.1 users/{uid} profile (typical fields)

Field	Notes
email, displayName, role	role is user for everyone except the two admin emails
profileType	grower adopter — immutable via client rules
growSetup, homeCity, growStyleNote	grower
adopterIntent, acceptedDevnet	adopter

chainOptIn, chainOptInAt	unlocks Tokenise / Market
solanaPubkey, walletVerified, walletProvider, walletLinkedAt	CF only
publicProfileEnabled, publicSlug, publicBio, profilePhoto	Stories / journal.growto.live
acceptedTerms, acceptedAge	signup

7.2 users/{uid}/app/state blob

```
{
  plants: [ /* see plant object */ ],
  entries: [ /* journal logs */ ],
  toolbox: { watering: [], feeding: [], environment: [] },
  journalSkill: { level, title, xp },
  coachProfile: { ... },
  plantWallet: { /* adopter tokens, optional */ },
  updatedAt: ISO-8601
}
```

Plant (ids are strings, often id-{ms}-{rand} or traffic ids). Keep these ids — seed mints and listings point at plantId.

id, name, strain, count	count = plants in a row/cohort
stage, subphase	stage keys above; subphase = pot / field ladder
startDate	grower-facing date; <i>not</i> used for market eligibility
createdAt, updatedAt	eligibility uses createdAt of plant or care logs
environmentName, environmentType	indoor / outdoor
fieldLocation, plantingLocation, exposureHours	outdoor
photo	Storage download URL (or legacy data URI)
notes, stageHistory, stageDates, subphaseHistory	

Entry

id, plantId, date	date is the journal day (YYYY-MM-DD). Market coverage ignores it.
type	opcenito zalijevanje gnojidba okolis presadjivanje stresori ostalo faza podfaza
note, photo, video, meta, createdAt, source	care days = distinct UTC days of zalijevanje/gnojidba <i>createdAt</i>

7.3 Top-level collections

Collection	Purpose	Who writes
users/{uid}/growerPosts	Stories drafts/published	owner
users/{uid}/notifications	in-app inbox	owner / buyer notify seller
users/{uid}/sharedGrants/{viewerUid}	admin shared library	owner / superadmin
publicJournalPosts	public Stories	Admin SDK (CF mirror)
publicGrowerProfiles / publicSlugClaims	journal.growto.live/g/{slug}	Admin SDK
marketListings	full listing (uid, sigs). Not a public board list.	owner create; buyer/seller status
marketPublicTape	scrubbed board (no Firebase uids)	Admin SDK only
seedMints / growthMints	NFT mint queues	owner creates pending; GH worker mints
platformRewards	test \$GROWTOO + adopter faucet	client request; worker pays
adoptStakes	stake mirrors	worker
harvestClaims	locked 50% release request	owner pending — fulfillment mocked
usedPaymentSignatures	replay guard	CF on listing write
aiUsage	daily Gemini caps	CF
loginEvents	admin report	signed-in self
opsHealthEvents	ops	privileged
plants / entries / tenants	legacy admin library	admin only — not the grower journal

7.4 Storage

users/{uid}/journal/{kind}-{stamp}-{rand}.jpg. Rules: owner read/write. Max ~5 MB after compression. If Storage fails, UI may fall back to inline data URI and must then respect the 1 MiB state-doc guard.

7.5 Market eligibility (keep in sync)

Duplicated in app/js/list-eligibility.js and functions/list-eligibility.js.

- Seed mint: not gated on coverage. Requires a real plantId in the grower's state.

- List / escrow activate: $\geq$ **14 elapsed days** since first care createdAt, and care on $\geq$ **50%** of min(elapsed, 180) distinct UTC days.
- Backdated entry.date does not count. Dumping 90 logs in one session is one care day.
- Listings with journalCoverage0k === false are hidden from the public tape.

8. Cloud Functions

All HTTP in europe-west1. Source: functions/index.js. Deploy: npm run deploy:functions.

Export	Role
healthCheck	public ping
solanaRpc	JSON-RPC proxy; strips faucet/admin methods
linkWallet	verify signature, set pubkey on user
coachChat	Gemini chat + optional image; verified email; quota 60/day
analyzeGrowFrames	1–2 JPEG frames; quota 25/day
kickChainQueues	dispatches GitHub chain-queues (PAT)
kickChainQueuesSchedule	every 2 minutes
settleMarketQueue (+ schedule)	sale_pending / cancel
reconcileMarketEscrow (+ schedule)	escrow_pending → active
verifyTxHealth	ops
onGrowerPostPublic / onGrowerProfilePublic	mirror Stories
onMarketListingHealth / onSeedMintHealth / onGrowthMintHealth	tape, coverage gate, queue kick, payment-sig reserve

HTTP ops (settle / reconcile / kick) require a Firebase ID token or GROWT00_OPS_SECRET.
Coach/vision/RPC/linkWallet send App Check tokens in monitor mode.

9. Solana / Devnet

Canonical addresses: chain/deployed.devnet.json and app/js/chain-config.js.

\$GROWTOO mint	3nReF8GGLdbPc4bmrgWyproVwt9taHb1yGvL5Cekrcqp · 9 decimals
Seed collection	79yYy4aSRzJQq9xonvcaTw7DgndoqwPMYDd2MpT8iVZa

Escrow program	GspPo6doBKoYmD6aCFHgo2q3CEXmWEoZXPpXAJnkjdyb
Marketplace PDA	2a887xGdhztkvfHn1BdR5xSkXjunzTG1StdTdtrRddAm
NFT escrow	EmQ4nNB1YVWNKVEiPNYhLgJR2gY1deJoV2L743z945yD
Care escrow (locked 50%)	C69K4V4921m1jYxjBoBoMYJR2fxYQpnx1w45gNGsL4ZU
Fee payer	Et1uJZn2GAWFdnKaVTubZYohKNJNB7gEpoQ7EHHKq975
Mint authority	F6ZEFk81ht6yWKvc5pLYQ5eM6DEKqdN69kbi2hFaMTv3 (also legacy escrow)

Queue path: app writes seedMints / growthMints / listing status → CF kicks GH Actions → chain/process-*.js with leased workers. Workflow: .github/workflows/chain-queues.yml (every 5 min + dispatch). Settlement mode program = atomic list/buy/cancel on the Anchor program; legacy hot-wallet path still exists for old listings.

Offer types: instant sale vs adopt_stake (50% now / 50% locked until harvest claim — claim pipeline not live fulfillment).

Traffic agents (chain/traffic/) are UX load personas with synthetic TrafficMint_ ids. Chain workers skip those. Do not confuse them with real Devnet mints. They can be wiped with npm run traffic:wipe.

10. App surface & modules

Primary nav (grower): **Journal • Log • Coach • Tokenise** (4th tab; Market is the other segment on that screen). Adopters see Market on that slot. view-dashboard and view-danas still exist in the DOM; showView redirects both to Journal. Tools/charts live under More as Measurements.

File	Owns
app/js/app.js	plants, entries, sync, weather, nav, toolbox, today, profile
ai-coach.js + coach-core.js	coach UI + local helper
plant-token.js	Tokenise, seals, wallet garden
market.js	board, invest, faucet, garden
escrow-program.js	Anchor list/buy/cancel from the browser
seed-chain.js	file mint requests
wallet-link.js / solana-wallet.js / solana-rpc.js	wallet + RPC
grow-camera.js / journal-photos.js	camera + Storage
grower-blog.js	Stories

list-eligibility.js	list gate (client)
notifications.js	inbox
signup.js	role payload
admin.js	allowlisted admin

Coach knowledge is curated in `functions/coach-knowledge.js` + journal snapshot from `coach-context.js`. It must not claim disease diagnosis or replace looking at the plant.

11. CI / secrets / ops

Netlify	Production = main. Rewrites: journal subdomain, /hr/*, /de/*, SPA 404. CSP is baseline only (wallets + Firebase inject).
GH: chain-queues	Secrets: FIREBASE_SERVICE_ACCOUNT_JSON, SOLANA_*_KEY_JSON, SOLANA_RPC_URL, dispatch token for kickChainQueues.
GH: chain-health, reconcile-market-escrow	scheduled health / escrow reconcile
GH: deploy-firestore-rules / indexes	on main
GH: deploy.yml	GitHub Pages mirror — excludes docs/ and functions/
Firebase secrets	GEMINI_API_KEY; ops GROWT00_OPS_SECRET; RPC URL for solanaRpc

Desk-test accounts (not traffic, not in git): grower test@growto.live; adopter adopter@growto.live; Luka traffic grower traffic+grower-s@growto.live. Passwords live in gitignored `chain/keys/desk-test-accounts.json` — do not paste them into this packet or a shared Drive copy.

12. What a rewrite must keep

1. **Stable Firebase UIDs** and `users/{uid}` profiles. Do not force a second signup.
2. **Plant ids** in the journal blob — they are foreign keys for `seedMints.plantId`, listings, and growth mints.
3. **Care semantics:** eligibility is `createdAt`, not journal date. Changing that silently rewrites who can list.
4. **Split the state blob** if you move to a framework: `plants / entries / toolbox` as subcollections, with a migration from the existing document. Watch the 1 MiB trap and photo URLs.
5. **Do not make profileType client-mutable** without a product decision. Two desks are a trust boundary (faucet, invest, Tokenise).
6. **Wallet link stays server-verified.** Never let the client set `solanaPubkey`.
7. **Public market board** reads `marketPublicTape`, not a list of `marketListings`.
8. **Devnet honesty:** no monetary value, no live harvest redemption, on-chain ≠ physical plant.

- 9. **i18n**: keep EN source + HR/DE; keep internal stage/entry keys; add DE to Gemini if the UI stays trilingual.
- 10. **Admin allowlist** stays email-based until you replace it deliberately.
- 11. **Chain workers and the Anchor program can stay** while the UI is rewritten. The fragile part is the 9k-line `app.js` + `localStorage` dual-write, not the mint queue.
- 12. **Move WeatherAPI off the client** on rewrite.

Suggested seams if you reshape rather than incinerate: keep Functions + chain queues + Firestore rules as the backend; replace `app/` with a typed SPA; migrate state; keep `locales/` key tree.

13. Existing public docs (data room already on the site)

Plant token / RWA architecture	/rwa-docs/
Risks (Devnet, no value)	/risks/
Terms / Privacy	/terms/ · /privacy/
Pitch chooser	/pitch/ (grower, adopter, investor, skeptic, labs)
Firestore rules notes	repo docs/FIRESTORE_RULES.md
App Check rollout	repo docs/app-check-rollout.md
Desk-test readiness	repo docs/TEST_READINESS.md (dated Aug 2026 — nav notes still useful)
i18n handbook	repo locales/README.md
Approved public claims	repo community-agent/config/product-facts.md (what marketing is allowed to say)

Generated from the live repo on 3 Sep 2026 for a rebuild kickoff. If a claim is not in this packet or in `product-facts.md`, treat it as unverified.